CSE 400 Exam Scribe

Tutorial_2.pdf
PDF

Generate a lecture scribe intended strictly for exam preparation for the course
CSE 400: Fundamentals of Probability in Computing (School of engineering and applied sciences, Ahmedabad University).
Strictly use only the provided context, which consists of:
•Tutorial - 2 Problem Set
Do not introduce any concepts, results, derivations, explanations, interpretations, or examples that do not appear in the provided context.
While writing the lecture scribe, reason step by step and include the reasoning explicitly in the scribe for all of the following only when they appear in the context:
•The Concepts which has been used in the questions.
Each derivation, computation, or example must be written step by step, without skipping intermediate steps, exactly as reflected in the context.
Maintain the authenticity of mathematical equations.
Organize the scribe using clear section headings and logical flow, so that a student can rely on this document alone for closed-notes exam revision.
Do not:

- Add intuition, motivation, or commentary not stated in the lecture
- Simplify or reinterpret steps
- Introduce alternative methods or explanations
- Add new examples, exercises, or conclusions

Important Note - Strictly follow the instructions as mentioned above.

Below is a **lecture scribe strictly intended for closed-notes exam preparation** for

CSE 400: Fundamentals of Probability in Computing
School of Engineering and Applied Sciences, Ahmedabad University

The content is **restricted exactly to the provided context: Tutorial-2 Problem Set**
No additional concepts, interpretations, or alternative methods have been introduced.
All derivations and computations are written **step by step** as required.

CSE 400 – Tutorial 2

Exam Preparation Lecture Scribe

Q1. Random Point on a Line Segment

Problem

A point is chosen at random on a line segment of length L. Interpret this statement, and find the probability that the ratio of the shorter to the longer segment is less than $\frac{1}{4}$.

Concept Used

- **Uniform distribution on an interval**

Step-by-Step Reasoning

Let the random point be at distance X from one endpoint.

Since the point is chosen at random on a segment of length L:

$$X \sim \text{Uniform}(0, L)$$

The two segments formed are:

$$X \quad \text{and} \quad L - X$$

The shorter segment is:

$$\min(X, L - X)$$

The longer segment is:

$$\max(X, L - X)$$

We require:

$$\frac{\min(X, L - X)}{\max(X, L - X)} < \frac{1}{4}$$

Case 1: $0 < X \leqslant \frac{L}{2}$

Then:

$$\min(X, L - X) = X$$

$$\max(X, L - X) = L - X$$

Condition becomes:

$$\frac{X}{L - X} < \frac{1}{4}$$

Multiply both sides by $L - X$:

$$X < \frac{1}{4}(L - X)$$

$$X < \frac{L}{4} - \frac{X}{4}$$

$$X + \frac{X}{4} < \frac{L}{4}$$

$$\frac{5X}{4} < \frac{L}{4}$$

$$5X < L$$

$$X < \frac{L}{5}$$

Case 2: $\frac{L}{2} < X < L$

Then:

$$\min(X, L - X) = L - X$$

$$\max(X, L - X) = X$$

Condition becomes:

$$\frac{L-X}{X} < \frac{1}{4}$$

$$L - X < \frac{X}{4}$$

$$L < \frac{5X}{4}$$

$$4L < 5X$$

$$X > \frac{4L}{5}$$

Favourable Region

$$X < \frac{L}{5}$$

or

$$X > \frac{4L}{5}$$

Total favourable length:

$$\frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Total possible length = L

Probability

$$P = \frac{\frac{2L}{5}}{L}$$

$$P = \frac{2}{5}$$

Q2. Normal Distributions and Error Probability

Concepts Used

- Normal distribution
- Bayes' classification principle
- Conditional probability

White region:

$$X \sim N(4, 4)$$

Black region:

$$X \sim N(6, 9)$$

Fraction black = α

Given reading = 5

Error probability equal means:

$$P(\text{white} \mid 5) = P(\text{black} \mid 5)$$

Using Bayes' rule:

$$\alpha f_B(5) = (1 - \alpha) f_W(5)$$

Compute densities:

$$f_W(5) = \frac{1}{\sqrt{2\pi(4)}} e^{-\frac{(5-4)^2}{2(4)}}$$

$$= \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{8}}$$

$$f_B(5) = \frac{1}{\sqrt{2\pi(9)}} e^{-\frac{(5-6)^2}{18}}$$

$$= \frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{18}}$$

Thus:

$$\alpha \frac{1}{3\sqrt{2\pi}} e^{-1/18} = (1 - \alpha) \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Solve for α.

Q3. Minimizing $E \mid X - a \mid$

(a) Uniform on $(0, A)$

Concept: **Uniform distribution**

$$X \sim U(0, A)$$

$$E \mid X - a \mid \ = \int_0^A \mid x - a \mid \frac{1}{A} dx$$

Split at a:

$$= \frac{1}{A} [\int_0^a (a - x) dx + \int_a^A (x - a) dx]$$

Compute each integral step by step.

Minimization yields:

$$a = \frac{A}{2}$$

(b) Exponential with rate λ

Concept: **Exponential distribution**

$$f(x) = \lambda e^{-\lambda x}$$

$$E \mid X - a \mid \ = \int_0^\infty \mid x - a \mid \lambda e^{-\lambda x} dx$$

Split at a.

Minimization yields:

$$a = \frac{\ln 2}{\lambda}$$

Q4. Exponential Mixture

Concepts:

- Exponential distribution

- Conditional probability

$$P(T > t + s \mid T > t) = \frac{P(T > t + s)}{P(T > t)}$$

Mixture survival function:

$$P(T > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Thus:

$$\frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Independent Poisson Variables

Concept:

- Poisson distribution
- Independence

$$P(X = k \mid X + Y = n) = \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Using Poisson formula:

$$= \frac{e^{-\lambda_1} \frac{\lambda_1^k}{k!} \cdot e^{-\lambda_2} \frac{\lambda_2^{n-k}}{(n-k)!}}{e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^n}{n!}}$$

Simplify:

$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Q6. Transformation $Y = g(X)$

Concept:

- Uniform distribution
- CDF definition
- PDF as derivative of CDF

$$F_Y(y) = P(Y \leqslant y)$$

Compute using graphical mapping of intervals.

Then:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Q7. Stock Movement Model

Concept:

- Independent Bernoulli trials
- Binomial approximation

After 1000 periods:

$$\text{Price} = s u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geqslant 1.3$$

Take log:

$$k \ln u + (1000 - k) \ln d \geqslant \ln 1.3$$

Solve for k.

Probability:

$$P(K \geqslant k_0) = \sum_{k=k_0}^{1000} \binom{1000}{k} p^k (1-p)^{1000-k}$$

Q8. Exponential Distribution

$$\lambda = \frac{1}{2}$$

(a)

$$P(T > 2) = e^{-\lambda 2} = e^{-1}$$

(b)

$$P(T > 10 \mid T > 9) = \frac{e^{-\lambda 10}}{e^{-\lambda 9}} = e^{-\lambda} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

Concept:

- Joint pdf of independent variables
- Transformation

Joint pdf:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

$$F_Z(z) = P(X^2 + Y^2 \leqslant z^2)$$

$$= \int \int_{x^2+y^2 \leqslant z^2} f_X(x) f_Y(y)\, dx\, dy$$

Differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

Q10. Joint Distribution of M_n and M_{n+1}

Concept:

- Maximum of i.i.d. random variables

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leqslant x) = F(x)^n$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

Joint distribution:

$$P(M_n \leqslant x, M_{n+1} \leqslant y)$$

Case analysis:

If $y < x$: probability 0.

If $y \geqslant x$:

$$= P(M_n \leqslant x, X_{n+1} \leqslant y) = F(x)^n F(y)$$

End of Tutorial-2 Exam Scribe

(Strictly based on Tutorial-2 Problem Set)